

Foundations in Deep Learning

Exercise session 0

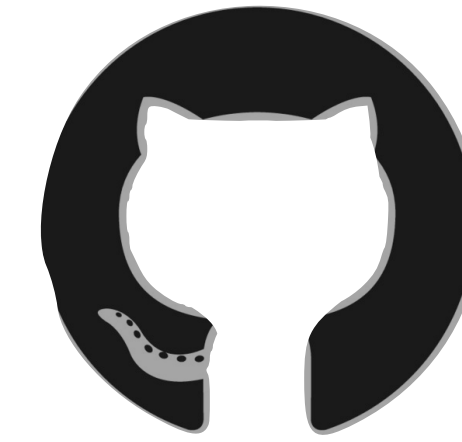
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Course organization



- **All teaching material will be posted on Github**
[https://github.com/alucchi/Foundations Deep Learning course HS26](https://github.com/alucchi/Foundations_Deep_Learning_course_HS26)
- **Forum on ADAM** to ask questions on course organization and course content (lectures and exercises)
 - Please use the forum or ask questions in person if it is about **course content**. For fairness reasons, I will not respond to private emails.
 - For any personal inquiries, you can of course reach out privately.

Organizational information

- **Grading** of this course is based on:
 - Continuous assessments (15%)
 - Midterm exam (29th October 2026, 16:15) (25%)
 - Final exam (4th February 2027, 10-12am) (35%)
 - Project (25%)

From the GitHub page:

Preliminary dates for continuous assessments (15% of final grade)

- CA1: Thursday, 22/10/2026 CA on Approximation and complexity theory
- CA2: Thursday, 29/10/2026 QA on Optimization
- CA3: Thursday, 19/11/2026 CA on Optimization Landscape in NN
- CA4: Thursday, 03/12/2026 QA on Architectures
- CA5: Thursday, 10/12/2026 QA on Transformers
- CA6: Thursday, 17/12/2026 CA on Generalization bounds I + II

Short survey on previous programming knowledge



<https://forms.gle/Gp8Y9xe64gu8AE8o7>

How the exercise sessions will go

- Weekly exercise problems, uploaded on Github repository
- Solutions are presented weekly by me or my colleague Nadezhda (probably mostly on the blackboard)
- Write-up of solution will be uploaded at the end of the session to ADAM (at least for first half of semester)
- Use forum on ADAM and exercise sessions to ask questions!

Exercise sessions are meant for you, not for me!

LLMs are phenomenal

Exercise 0: Prerequisites – Solutions

Foundations of Deep Learning, Fall 2026

Problem 1: Operator Norm

Let $A \in \mathbb{R}^{m \times n}$, with $m \geq n$, and

$$\|A\|_2 := \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2}.$$

(a) **Definition of a norm and verification for the operator norm.**

A norm on a real vector space V is a map $\|\cdot\| : V \rightarrow [0, \infty)$ such that for all $u, v \in V$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned} \|u\| &\geq 0, & \|u\| &= 0 \iff u = 0, \\ \|\alpha u\| &= |\alpha| \|u\|, \\ \|u + v\| &\leq \|u\| + \|v\|. \end{aligned}$$

We verify these properties on $V = \mathbb{R}^{m \times n}$.

Non-negativity and definiteness. Since the Euclidean norm is non-negative, $\|A\|_2 \geq 0$. If $A = 0$, clearly $\|A\|_2 = 0$. Conversely, if $\|A\|_2 = 0$, then

$$\|Ax\|_2 \leq \|A\|_2 \|x\|_2 = 0$$

for every x , hence $Ax = 0$ for all $x \in \mathbb{R}^n$. Therefore $A = 0$.

Absolute homogeneity. For $\alpha \in \mathbb{R}$,

$$\|\alpha A\|_2 = \sup_{x \neq 0} \frac{\|\alpha Ax\|_2}{\|x\|_2} = |\alpha| \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = |\alpha| \|A\|_2.$$

Triangle inequality. For any $x \neq 0$,

This is what you'll get from me

Problem 1 (Operator Norm):
Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a rectangular matrix with $m \geq n$.

a) Recall the definition of the operator norm of $\mathbf{A}$:

$$\|\mathbf{A}\|_2 = \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}x\|}{\|x\|}.$$

Write down the definition of a vector norm. Show that the operator norm is indeed a vector norm in the vector space $\mathbb{R}^{m \times n}$ of matrices.

b) By the singular value decomposition (SVD) of $\mathbf{A}$, show that $\|\mathbf{A}\|_2 = s_1(\mathbf{A})$, the largest singular value of $\mathbf{A}$.

c) By SVD of $\mathbf{A}$ again, show that $s_1(\mathbf{A}) = \sqrt{\lambda_1(\mathbf{A}^T \mathbf{A})}$, where $\lambda_1(\cdot)$ denotes the largest eigenvalue.

if you wonder about the difference on supremum and maximum, this post explains it nicely!

<https://math.stackexchange.com/questions/160451/difference-between-supremum-and-maximum>

a) Let $\|\cdot\|$ denote an arbitrary norm on some space S , e.g. $\mathbb{R}^n$. Then it is characterized by the following properties:

- (i) non-negativity: $\forall x \in \mathbb{R}^n: \|x\| \geq 0$; $\|x\| = 0 \iff x = 0$
- (ii) absolute homogeneity: $\forall \alpha \in \mathbb{R}, \forall x \in \mathbb{R}^n: \|\alpha x\| = |\alpha| \cdot \|x\|$
- (iii) triangle inequality: $\forall x, y \in \mathbb{R}^n: \|x+y\| \leq \|x\| + \|y\|$

Examples: Vector norms:

- Euclidean norm: $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$
- Maximum norm: $\|x\|_\infty = \max_i |x_i|$
- L_1 norm: $\|x\|_1 = \sum_{i=1}^n |x_i|$
- L_p norm: $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$

Let $A \in \mathbb{R}^{m \times n}, m \geq n$.
Show that $\|A\|_2 := \sup_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\|Ax\|_2}{\|x\|_2}$ is a norm.

(i) non-negativity: $\forall A \in \mathbb{R}^{m \times n}: \|A\|_2 \geq 0, \|A\|_2 = 0 \iff A = 0$